

Abstracts Update 24th of June

Student: Roos Geerse

Location: Utrecht University (Montoya)

Type of project: internship

Changing the narrative about testosterone deficiency in aging women.

Testosterone deficiency in aging women can cause sexual and other psychosocial problems, but this is still not an accepted idea, partly because the underlying mechanism is poorly understood. As a result women may be offered psychosocial counseling that will have little effect, because their problems have a physical cause. To help solve this problem I have proposed a model of emotion and cognition that can explain the psychosocial effects of testosterone and the sexual response cycle. In my internship I have worked further on this model and applied it to testosterone deficiency in aging women. In addition, I made a plan to disseminate my conclusions so practitioners in the Netherlands could make an informed decision on diagnosis and treatment.

Student: Hanna Sundermeier

Location: Utrecht University (Terburg)

Type of project: thesis

Postauricular muscle reflex as a scalable indicator of positive affect

Despite being a vestigial muscle, the postauricular muscle reflex has shown to be an objective indicator for subjective reward experience. The study at hand provides a test protocol to examine whether the postauricular reflex is scalable. If the intensity of the reflex varies and does indeed reflect the degree to which one is experiencing positive affect, then this will have meaningful implications for assessment and research of various psychological disorders. The scalability can be tested by following the steps in the protocol at hand which combine EMG measurements, rewarding stimuli, personality assessments and instructions for calculations to evaluate the results.

Student: Felice de Schutter

Location: BeAnotherLab (Barcelona)

Type of project: thesis

Worldwide, people suffer from depression and next to their health condition, they have to deal with stigmatization. Interpersonal contact, perspective taking, and empathy have shown to play important roles in making anti-stigma treatments effective. Virtual reality (VR) can promote these aspects. With the aim to determine whether VR should be used in depression stigma programs, this study examined how prior contact to people with depression, baseline empathy and perspective taking stand in relation to depression stigma. By means of an online survey, data from 203 participants was collected. Results indicate that prior contact, being able to take the perspective of a person with depression, as well as perceived similarity to the stigmatized group play important roles in depression stigma. With its ability to put the user into the shoes of a person with depression, thereby

facilitating perspective taking and promoting in-group perception, we suggest VR to be an effective tool for depression stigma reduction.

Student: Carlijn van de Pas

Location: GGZ Centraal

Type of project: thesis

Patients with BPD are found to interpret ambiguous faces as more negative and threatening (interpretation bias) and have trouble disengaging their attention away from threatening faces (attention bias). To improve this emotional dysregulation within BPD patients, GGZCentraal has developed a training tool helping them making their biases more positive. This tool is still in development and therefore, aim of present research is to test the effectivity and usability of this tool in a non-clinical sample. During a 7-days experiment, participants are asked to complete some questionnaires and tasks in the tool. In the end results from pre-tests will be compared with results from post-tests to check whether the tool is effective in its goal. Moreover, usability questionnaires will be conducted to check for any usability problems. Goal is to improve the usability and with that the effectivity and make testing and eventually implementing on BPD patients possible.

Student: Ditte van Haalen

Location: Utrecht University (Keizer)

Type of project: thesis

The role of interoception and emotion regulation in patients with severe early childhood trauma and borderline personality disorder.

The interoceptive ability to perceive the body's signals is thought to play an important underlying role in emotion regulation. The current study therefore investigated how affective touch, a form of interoception with social significance, is perceived in two patient groups that are characterised by a distorted emotion regulation and disconnection to their body, namely borderline personality disorder (BPD) patients and patients that have experienced severe early childhood trauma. The purpose of this study was twofold: first it was investigated if and how these patient groups differ in their perceived pleasantness of affective (3cm/s) and non-affective touch (30 cm/s), secondly it was explored whether touch was associated with emotion regulation in the three different groups. We found that both patient groups showed lower perceived pleasantness of affective touch compared to healthy controls, but no differences were found for non-affective touch. Furthermore, BPD patients showed less extreme differences in pleasantness ratings between affective and non-affective touch compared to healthy controls and trauma patients perceived affective and non-affective touch as equally pleasant. In the trauma group, we found no relationship between either form of touch with emotion regulation, suggesting that these patients might not profit from the benefits associated with affective touch in the same way as healthy people do. In the BPD group, we found a relationship with emotion regulation for both affective and non-affective touch, suggesting that these patients might have a different relationship with touch compared to healthy controls, in which we found that only affective touch can predict emotion regulation problems.

Student: Lena Siegling
Location: TNO
Type of project: internship

I am currently working on my thesis in collaboration with TNO. As costs of artificial intelligence (AI) continue to decrease, and efficiency of this technology increases, AI is rapidly being adopted across domains. A critical factor to consider for productive and safe human-AI collaboration is user-trust. Previous research suggests that communicating system uncertainty can help to calibrate appropriate levels of trust. To investigate this within robotics, I have designed a 2x2 mixed factorial analysis during which participants interact with an AI drone that guides them through an abandoned house. The drone makes a mistake and sometimes indicates its uncertainty or apologizes for its faulty advice. I am using questionnaires to investigate effects of different drone communication on participants' trust at different instances during collaboration. The experiment is distributed via Qualtrics. I am hoping to finish data collection in the first week of July.

Student: Pim Verhagen
Location: TNO Soesterberg
Type of project: thesis

The remote control of robots (telerobotics) generally requires a high level of expertise and may impose a considerable cognitive burden on operators. Ideally, operators should have the (illusory) feeling that the robot's arm and hands are their own, so that they do not notice the operation is being mediated. This feeling is often referred to as the sense of embodiment. In this study we aimed to validate the cross-modal congruency effect (CCE) as an objective measure of embodiment in four teleoperation conditions. We found that the varying degree of realism in the conditions affected the magnitude of the CCE. However, comparisons with subjective reports indicate that the CCE might not be a robust measure of embodiment strength.

Keywords: embodiment, teleoperation, cross-modal congruency effect

Student: Emma van Gruijthuijsen
Location: Achmea
Type of project: thesis

Nudging towards Honesty in Insurance Fraud

This study aimed to improve honest claiming behavior in Achmea's Occupational Disability Insurance. Two nudges were emailed to clients in this field experiment: a Deterrence nudge and a Contact nudge. Clients of Achmea were asked to respond with a percentage. The two measures of honest claiming behavior are the Response Rate as well as the Average Percentage Increase. The results are outstanding: the two nudges on average performed 8.75% better than the control group in terms of Response Rate (Wald test; $n = 386$; $p = 0.018$). As for Percentage Increase, the Contact nudge and the control group outperformed

the Deterrence nudge on average by 5.81% (t-test; $n = 386$; $p < 0.001$). Therefore, the Contact nudge performs best. Compared to nudging fraud literature, the Contact nudge almost doubles the average results.

Student: Heike Diepeveen

Location: NLR (Dutch Aerospace Centre), Amsterdam

Type of project: combined internship & thesis

Fatigue is an important factor in aviation accidents and incidents and cannot always be prevented, due to long flights or jet lag. As of today, there is no quick, non-invasive method to assess fatigue in an aviation setting, such as a cockpit or air traffic control tower.

The aim of my project is to investigate whether voice intensity and pitch change significantly when someone is more fatigued. If this is the case, a voice analysis system can be integrated with speech recognition systems present in cockpits and control towers. This way, fatigue can be detected in real time and direct measures can be taken.

I am executing an online study in which people have to do a reaction time task, fill in questionnaires and do two speech tasks. One speech task is aimed at recording everyday speech, while the other task is aimed at recording procedural speech. For both speech tasks, I will compare two conditions within subjects. In one condition participants slept 8 hours or more and in the other condition 6 hours or less.

My hypothesis is that voice pitch and intensity have significantly lower values in the sleep deprived condition. Results will follow in August.

Student: Annick Wagemans

Location: Visio

Type of project: thesis

My thesis is about eye-movement disturbances in patients with acquired brain injury. These ABI patients are reporting "Non Specific Complaints" like dark haze, blurry vision, increased annoyance by light sources. So more specifically, I will be looking whether these Non Specific Complaints(NSCs) affect the performance on the three eye tracking tasks I have conducted. These three tasks are suitable to measure performance on the three main types of eye movement: saccades, fixation and smooth pursuit.

Hopefully, after my analysis, it will be clear whether or not some of these NSCs affect the performance on some of these tasks.

Student: Tommaso Ceroni

Location : Bloakes

Type of project: internship

PRE-TESTING NUDGES

Since the publication of the book *Nudge: Improving Decisions about Health, Wealth and Happiness* by Richard Thaler and Cass Sunstein it became more clear the potential that governments have to shape people's behavior in a more innovative, rather than coercive

way through nudges. The design of nudges requires a money investment, therefore failing in changing behavior will most of the time result in money lost. Being able to test in advance nudges would prevent wasting money in bad designs. This review through the explanation of nudging, digital nudging and their implementation in contexts such as the COVID-19 pandemic, has the objective to understand whether or not it is possible to pre-test nudges with implicit tests. This dissertation focused on both positive aspects and critics about nudging and IRTs, by also inserting solutions that academic researchers provided to solve the limitations of these tools.

Student: Aafke van Welbergen

Location: TNO

Type of project: thesis

Influence of depth cues on cybersickness

The simulation of a (non-)realistic world with Virtual Reality (VR) is very popular. However, VR also has some drawbacks. A big concern is the causation of cybersickness, a form of visually induced motion sickness which is caused by a conflict between the visual and vestibular system. To prevent cybersickness, this mismatch needs to be reduced. This reduction can be established in several ways, depending on the presence of actual self-motion.

This study aimed to show the influence of depth cues on cybersickness in relation to self-motion. Due to extraordinary circumstances, only a pilot study was conducted, limited to test the influence of motion parallax in situations with actual self-motion. The results show no significant effect of motion parallax on cybersickness. However, the individual data indicates that cybersickness might increase faster without motion parallax. Further research with a larger sample size and all conditions is necessary to gain more insight.

Student: Zhaoxi Luo

Location project: Utrecht University (Dijkerman)

Type of project: internship

Abstract In our lab, there is a very interesting machine called the MIRAGE system. Participants will look at images in real-time of their limbs from the same perspective like viewing the true body parts by using mirrors and cameras. Put a part of your body into the MIRAGE system, you can see their shape and size change, and you will still believe it is a part of your body. In my experiment, I want to investigate whether distorting the size of the hand palm in the MIRAGE system will affect time perception. I derived the participant's perceptual threshold using the Temporal-Order Judgment Task(TOJ) task in the original size of hand and the distorted size of hand in the MIRAGE system respectively. The TOJ task is one of several psychophysical paradigms used to measure the efficiency of temporal resolution in this time domain. It reflects the ability to perceive the temporal order of (at least two) stimuli presented in rapid succession. By comparing the differences in the data obtained in this task, I will draw my final conclusion.

Student: Julia Meijer
Location project: Rijkswaterstaat
Type of project: internship

A literature study on human factors of the current jam tail support system and a possible future in-vehicle jam tail support system.

Abstract: The conventional driving task is changing to a collaboration of the driver and the car, because of this driving has become more complicated for drivers. The driver has to process more information and some tasks are taken over by the car. One of the areas in which change is happening right now is real time information on upcoming jam tails. Drivers are given more information for this via nomadic devices and have to elicit certain behavior to avoid collisions. Getting more information while driving is an added challenge to the task. Certain aspects on human factors have been reviewed to properly give this information. Perceiving (e.g. adding auditory and/or tactile stimuli) and comprehending of information (e.g. only alerting or also adding maximum speeds), personalization, homogeneity and age (e.g. elderly people) will be discussed to create an advice on what should and should not be used for in-vehicle jam tail support systems. The most important results show that auditory cueing by human speech has preference. Also, all other information should be cut off when receiving information and shortly after. Lastly homogeneity is crucial and incongruent information should be avoided.

Student: Isabela Assiria de Oliveira Bossi
Location: TNO
Type of project: internship

The goal of this internship was to explore how to use data visualization to facilitate the work of a real-time intelligence agent of the Emergency Services of the Netherlands (112-meldkamer). This agent is the person responsible for alerting the appropriate emergency service (police, fire brigade and ambulance) when a phone call comes in through 112. The method within data visualization chosen for this project is called Network Visualization, which is a way of displaying relationships between elements of a network. By researching this method I was able to design and prototype a custom solution to make the display of the data simpler and more efficient for the agent, speeding up this highly important task and eventually saving lives.

Student: Daniela Pistone
Location: Unravel research
Type of project: thesis

Predicting hit songs: Using neural synchronicity on songs to predict Spotify plays.

Predicting hit songs is mostly done by data analysts use big data and algorithms, but neuroscience likewise tried to research predictive measures too. It is found that when humans are experiencing the same stimuli, brains behave alike. This paper researches if early stage popularity on Spotify can be predicted by this neural synchronicity. An experiment was conducted with two albums that just got released. The albums were made

into smaller fragments and participants listened to the fragments while wearing an EEG. We found that the subjective likability ratings of the songs were not predictive of Spotify plays. However, the neural synchronicity metric was found to significantly predicted the Spotify plays. These results suggest that neural synchronicity could be a valuable metric for predicting hit songs. The metric could help the music industry with the decisions regarding choosing songs for an album or single.

Student: Martijn van Hattum

Location: Validators

Type of project: internship

Emotions are getting increasingly important in marketing and advertising due to the effectiveness of emotional engagement in these fields. Therefore, it is important to accurately measure emotion. Two of the most implemented neuromarketing tools to measure emotion are EEG and fMRI. The market research company Validators implement two in-house designed methods to measure emotion: an Automated Facial Coding(AFC) tool called Emotionflow and a moment-to-moment measurement tool Valueflow. The purpose of this explorative research is to investigate how Validators can create competitive advantage on their methods(Emotionflow and Valueflow) compared to EEG and fMRI. To do this, the first part of this research investigated the way these four methods measure emotion. In order to place these methods in a commercial context, the second part compared the four methods on various commercially relevant aspects.

Student: Vani Pillay

Location: Music Brain Health Technology Lab at Leiden University

Type of project: thesis

Covering the Cost of Paying Attention

Children with attention deficit/hyperactivity disorder (AD/HD) struggle in normative education environments. There are drawbacks to medication treatment due to the side-effects, cost and potential nonadherence. Research has also shown that mood may influence selective attention, with positive mood having a negative effect on attention and negative mood having a positive effect on attention. Because of this, alternative attention training methods are being developed. Music therapy has been found to help through the implementation of music-based interventions. The aim of this research is to determine if mood should be considered when creating an online music-based intervention task for children with AD/HD. The experiment was conducted using Qualtrics, where mood was assessed using the self assessment manikin and selective attention was assessed through an omitted note music task, where participants had to identify missing musical notes in different pieces of music. Results of the experiment are not yet available.

Student: Dayenne Teulings

Location: TNO Soesterberg

Type of project: thesis

The effect of eye position on the Oculogyral Illusion

The aim of the current study was to test what the effect of eye position was on the Oculogyral Illusion (OGI). In addition, it was tested whether this effect depends on rotational speed, and if there was a difference in acceleration/deceleration and clockwise/counter clockwise. Since literature shows that the underlying mechanism of the OGI equals vestibular diseases like Ménière Diseases (MD), it was tested whether the effect of prisms (as treatment for MD) reduces the OGI the same way it reduces the symptoms of MD. It was hypothesized that a side eye position, similar to prisms, should reduce the OGI. 24 subjects ($M_{\text{age}} = 34.54$, $SD_{\text{age}} = 13.042$) participated in a single experiment on a rotation chair, in a complete dark environment, with slow horizontal angular acceleration, focussing on a chair-fixed target. The results of a three-way ANOVA showed that a side eye position indeed reduces the OGI, and that both rotation speed and rotation direction has an significant effect.

Student: Molly Adams

Location: Utrecht University (Wachner/de Ridder)

Type of project: thesis

Nudging is a popular but controversial means of influencing behaviour. Although nudges should always benefit the individual decision maker and/or society, there is concern regarding the ethicality of using behavioural insights to influence behaviour. Nudge transparency has been proposed to protect decision makers from manipulation, with research showing that disclosing the presence of a nudge does not reduce nudge effectiveness. Findings from Paunov, Wänke, and Vogel (2018; 2019) suggest that nudge disclosure may even enhance nudge effects, but this result has not been replicated. The aim of the present research was to replicate and extend the results from Paunov and colleagues (2018; 2019). Three conditions were assessed in an online experiment: a transparent default, a non-transparent default, and a control condition. Additionally, measures of participants' feelings towards the nudge and the researchers were assessed to understand the mechanisms associated with their decisions.

Student: Rosemarijn Damen

Location: Utrecht University (Baas)

Type of project: internship

In this internship I have worked on an online fear conditioning experiment at Utrecht University. Studies have found different types of conditioners in fear conditioning experiments, which are associated with anxiety disorders and treatment failure of these disorders. These studies have found certain characteristics associated with the different types of conditioners as well, such as trait anxiety and hormone levels. I will replicate these findings, and additionally will investigate fear of the coronavirus and whether that can be associated with the different types of conditioners. To replicate these findings in an online research, I have set up an online experiment in Gorilla. As of right now, I have already started gathering data. After collection of the data, I will analyse the data and write a report on it, and I will write a data analysis pipeline document.

Student: Lars Dijkstra
Location: Unravel Research
Type of project: thesis

In this eye tracking study we aim to understand the effect of mind wandering and attentional lapsing on commercial processing and ad effectiveness. We exposed our participants to a varying number of commercials and aimed to identify different viewing patterns as a result of serial positioning, logo placement and brand popularity. Due to the cognitively demanding nature of commercial processing, we expect our participants to show different viewing behaviour as a result of fatigue and cognitive saturation. If viewing behaviour is indeed impacted by serial positioning and fatigue, we should question the effectiveness of traditional media buying and tv ad placement all together.

Student: Nicole Plungjan
Location: Signify
Type of project: internship

What constitutes User Experience Research in the real (corporate) world? When there are deadlines to keep in mind, a boss to keep happy and company morals to uphold? I took a deep dive into a corporate world that simultaneously produces high tech toys for consumers, incredible technology for the agriculture sector and overarching systems for the retail sector. This presentation is about my internship at Signify, formally known as Philips Lighting. I took part in multiple projects, from start to finish, most as the Lead UX researcher in that project's team.

In this presentation I will talk about the different UX research strategies I got to employ at Signify, and how we managed to still continue the research while in the middle of a global pandemic.

Student: Helen Visser
Location: Wageningen University Research
Type of project: thesis

The current demand for food exploits the resources of the earth. In order to reduce the greenhouse emissions, people should transition to a more plant-based diet. This gives reason to explore the contextual influences on food habits as research shows that context can make a difference for the maintenance of new habits.

In the current experiment, 91 participants adjusted their breakfast habits for four weeks by either adding an apple or swapping their dairy product for a plant-based product. Weekly habit strength measures were compared for two groups i.e. 'routinized' and 'less routinized' for 5 weeks (with a baseline week included). A significant interaction between habit strength over time and context stability (CS) was found, $F_{(3, 91)} = 3.365$, $p = .019$, $\eta^2 = .036$, implying that people with a higher CS are more resilient in habit strength adjustment.

Student: Jan van den Heijkant
Location: Utrecht University (Janssen)
Type of project: internship

For my internship assignment I will be working with Chris Janssen at Utrecht University. He has been working on a project to try and inform the public about (his) research findings through YouTube videos. Chris's work mostly deals with automation and multitasking. Some of the topics I can choose from his recent research are:

- videocalls using augmented reality during driving tasks
- the influence of cognitive load on susceptibility to audio
- Integrating knowledge of multitasking and interruptions across different perspectives and research methods.

I will be making one of these videos about Chris's research as part of my internship assignment. I hope to start this project next week (June 15th) and plan to finish it around the end of August.

At the moment I haven't picked a specific topic yet as my thesis ran slightly longer than planned, but at the time of the update day I hope to have a clearer plan.

Student: Ya Liu
Location: Perfect World (Beijing, China)
Type of project: Internship

I'm doing my internship in a company that is focused on outputting cultural content. I'm going to output content about psychology, including publishing a book about psychology and developing a mini-program about psychology on WeChat (a social APP in China). The psychology book is aiming to combine some psychology academic research with Chinese history and our daily life. The WeChat mini program we are going to develop is based on an IP called "Panorama of human", which is a book, published by our company. We hope to present the content of this book on this mini-program, and I'm going to design some questionnaires to test the user's mental state. Besides, I'll work on the Interactive design of this mini program.

Student: Claire Borst
Location: Bloakes
Type of project: Internship

According to the Theory of Planned Behaviour consumer attitudes affect consumer behaviour intentions (Ajzen, 1991). This theory is taken into account to provide insights in the effectiveness of an Evaluative Priming Test in predicting buying preference for magazines by implicit associations on emotional states and intention to buy. This is done by implicit measures, because the acquired impulsive reactions are of better influence in driving behaviour than explicit measures (Fazio et al., 1986). Current study used implicit reaction time scores from two Evaluative Priming Tasks, which is used to assess evaluative

responses to any object that is presented as a prime-stimulus in the priming test (Gawronski, 2014). The Forced Choice Priming Task is a speeded judgement task in which participants have to categorize targets in the corresponding category after being primed with several magazine covers. In the Free Choice Priming task participants were free to categorize different magazine covers into the categories buying or not buying. Results showed that associations between emotions and magazines are stronger than associations between intentions to buy and magazines. More specifically, positive words have been shown to evoke stronger implicit associations for magazines than negative words, regardless explicit affinity. However, when predicting buying preferences for magazines, intention to buy is a more accurate parameter than emotional states. This study indicated that an Evaluative Priming Test is effective in predicting buying preference for magazines by implicit associations on emotional states and intentions to buy.

Student: Kaja Sramova

Location project: Trimbos Instituut (Utrecht)

Type of project: internship

RECOVER-E SharePoint

RECOVER-E is a multinational project striving to implement community-based mental health care for people with severe mental health problems in South-Eastern Europe. Microsoft SharePoint was selected to serve as a management information system for this project, facilitating communication and document-sharing between the project partners. However, the platform remained largely unused. The aim of the internship assignment was to investigate the barriers and facilitators of the SharePoint functionality in RECOVER-E. Thematic analysis of an interview with a member of project coordination and of a survey with the project partners pointed to the main reasons behind the unsuccessful implementation of SharePoint in practice – insufficient time, different priorities, lack of adequate personnel – but also indicated that SharePoint was important for the project nonetheless.

Student: Sietske Bootsma

Location: Utrecht University (van Maanen)

Type of project: thesis

Driving behavior and mental workload

Driving tasks are cognitive behavioral processes involving information perception, decision-making, and execution. Driving is a very common but highly complex task that involves dynamic interleaving and execution of multiple critical subtasks. Therefore, understanding driver intention and behaviors are beneficial to driver safety, vehicle drivability and traffic efficient. The complexity of all the different sub processes that take place during driving is widely studied to gain a better understanding of driving behavior. A specific widely studied concept is ‘multitasking’. When a driver has to perform more tasks at the same time this can be very demanding and a higher mental workload can be the consequence. A specific form of driving is lane changing. This paper builds upon the existing paper of van der Heiden.

Janssen, Donker and Merckx (2019) and examines which components of mental workload are responsible for the decrease in drive performance during lane changing.

Student: Yuhua Zhang

Location: Utrecht University (Beckers) / Bloakes

Type of project: internship

IAT, implicit association test was first developed to address the implicit attitudes in social cognitions. And now researchers are increasingly interested in its applications in predicting consumer behavior. In my internship, I will do a literature study on this topic and present the review in a practical manner, which means it can be used as a guideline for researchers in companies. In the first part, all important considerations including the general procedure, the scoring algorithm, the strategy of using IAT are fully discussed to provide systematic knowledge of conducting IAT in consumer study. In the second part, some specific examples from literature are reviewed and the question of representativeness is discussed for each case. In the end, some useful insights from literature are provided. Through the literature study, I hope the IAT can be applied in consumer studies like regular questionnaires and this review will serve as the ideal protocol.

Student: Anaïs Thijssen

Location: Utrecht University (Baas)

Type of project: thesis

For my thesis I am investigating whether the use of cannabidiol could help patients with treatment resistant anxiety disorders. Cannabidiol is a non-psychotropic component of cannabis that has little side effects and shows no tolerance or withdrawal symptoms. An often used treatment for anxiety disorders is exposure therapy, where patients learn that the thing they fear is in fact not followed by the anticipated aversive event. This learning process has been proposed to work through the endocannabinoid system (ECS). Treatment resistance therefore might be due to hypoactivation of the ECS. Since cannabidiol is able to activate the ECS, we tested in a randomized double-blind placebo controlled experimental trial whether the use of cannabidiol during exposure would facilitate the extinction of the maladaptive fear and thereby provide an effective treatment for these patients. It is still work in process, so no results yet.

Student: Carolin Henze

Location: Metrixlab

Type of project: internship

MetrixLab is a global market research company operating in over 90 countries worldwide. Their goal is to provide clients with valuable insights into brand engagement, customer value, and product innovation. Among their clients are renowned companies such as Heineken, Philips, and Google. Whenever MetrixLab carries out research, all findings need to be put into a presentable deliverable. This can, for example, be a PowerPoint

presentation, a written report, or a scorecard using an online dashboard. Scorecards are graphs that visualize the information in a concise and easily understandable manner, which can be presented to the client. As soon as data is entered into the dashboards all online scorecards are automatically updated. There is no need to change a report or make a whole new PowerPoint presentation which saves a lot of time and sweat. However, a lot of researchers and data analysts at MetrixLab are hesitant about using the dashboard or have trouble using it. Building on previous internship projects, my goal is to reduce the need for support to make the whole process faster and more efficient. Through usability interviews, I want to find out what can be done to improve dashboard functioning.

Student: Julia van der Donk
Location: Effectory in Amsterdam
Type of project: thesis

Happy employees are considered one of the pillar stones of a happy and healthy organization. Research on this subject is called people analytics and is a data-driven approach to managing people at work. Therefore, this is considered a hot topic for organizations in order to be a successful organization. This research uses data collected by Effectory, the number one independent employee research company in the Netherlands, and consists of data over 100.000 respondents in the Dutch Healthcare-sector. During the explorative phase the variables workload, support, collaboration, happiness, eNPS and retention are selected. This phase focuses on selecting items and discovering probabilities, outliers, reliabilities and inconsistencies. A multiple regression analysis using the final data results in a model that predicts and shows the connections and importance of the variables and more detailed how the variables may lead to retention and eNPS. As a result, this research can advise organizations on achieving employee happiness and how to make the working world a better place.

Student: Giselle Veerman
Location: Utrecht University (Janssen)
Type of project: Thesis

During my thesis at Utrecht University, I investigated the effect of solely cognitive load on auditory susceptibility using an auditory verb generation task. In this task, participants heard a noun and were instructed to generate a verb related to the noun (e.g. “apple” and “eating”). In addition, I investigated the effect of actively responding to auditory signals on auditory susceptibility, by requiring half of the participants to give a response after hearing a certain sound. To measure susceptibility, I recorded the frontal P3 peak in the brain activity using electroencephalography. The results show convincing evidence for a reduced frontal P3 (and therefore auditory susceptibility) during cognitive load conditions. However, there’s insufficient evidence to conclude whether response requirement can overcome this reduction. In conclusion, these results suggest auditory susceptibility can diminish due to solely cognitive load. Further research is required to obtain sufficient evidence regarding the potential effect of active responses.

Student: Alieke Struik
Location: Delft University of Technology
Type of project: thesis

Meaningful human control over automated driving systems, driver intentions and ADS behaviour.

Automated driving systems (ADS) are the future of driving, but there are concerns about responsibility and control over these systems. The philosophical concept of meaningful human control (MHC) was applied to ADS to answer these questions. One of the conditions of MHC requires that the vehicle takes into account the intentions of at least one human, which can also be someone else than the driver. The research question that follows is: what differences between driver intentions and ADS behaviour do drivers accept? To investigate this, the intentions and preferences regarding driving behaviour of 220 participants will be measured in an online experiment. Their intentions regarding driving behaviour will be measured first. Thereafter, they read vignettes which describe situations where an ADS initiates or suggest an action and the participant can choose to adhere to the ADS' decision or take-over control/ ignore the suggestion of the ADS. Data will be analysed to find out whether people accept an ADS being in control and deviating from their intentions.

Student: Niels van Kuijk
Location: Visio
Type of project: thesis

Patients with visual field loss (VFL) can experience difficulties in their mobility. Occupational therapists do not always have the needed information from VFL patients to help them with their problems. Recently, a company in the Netherlands (Royal Visio) started using eye trackers during mobility lessons of VFL patients to collect this needed information. This thesis project created one e-learning to make eye trackers more user-friendly for occupational therapists, and did one experiment with occupational therapists and object recognition software YOLOv3. This goal of this experiment is to learn what occupational therapists and YOLOv3 see as relevant street elements during a mobility lesson with a VFL patient. This is examined in order to increase the effectiveness of the use of eye tracker videos by occupational therapists. The corresponding research question of this experiment is: In what degree are occupational therapists and object recognition software YOLOv3 able to analyse an eye tracker video correctly, whereby occupational therapists can draw correct conclusions on what the needed information is for a specific patient?

Student: Elles van Timmeren
Location: Utrecht University (informatics, de Graaf)
Type of project: thesis

Differences in behavior in HRI as opposed to human-human interactions are becoming more and more important as humanoid robots will become more integrated in our society.

Previous research on prosocial behavior and emotions in HRI is inconsistent. Therefore, I investigated how differences in feelings of guilt relate to differences in prosocial behavior in HRI and HHI.

The experiment was originally meant to be in a real-world setting, however, due to COVID-19 I had to make an online version of the experiment. Participants played a guessing game against a robot or a human, a version of the Prisoner's dilemma and filled in some questionnaires relating to confounding variables as empathy and likability. The results are being analyzed right now and will hopefully give insights in the behaviors, and potentially differences, people have towards robots and humans and the feelings underlying those behaviors. The results will hopefully help governments and policy makers.

Student: Tiemen Wagenvoort

Location: Utrecht University (Benjamins)

Type of project: Thesis (combined with internship)

Studying gaze behavior using eye tracking can give insights in decision making processes regarding food choices. However, commonly used analyses that involve area of interests (AOI's) are often problematic in complex environments like a supermarket. In these cases it could be beneficial to use general outcome measures that do not use AOI's. In this study, we investigate with wearable eye tracking if these measures can account for differences in gaze behavior for customers that already decided what to buy (resembling a search task) or did not yet decided what to buy (resembling a decision task). Additionally, we take into account their familiarity with the supermarket (resembling learning effects). This study substantiates the Natural Decision Segmentation Model by providing evidence for differences in gaze behavior regarding the 'orientation' and 'comparison' phase using general outcome measures. Although the results should be considered with nuance, they could support the application of wearable eye tracking research in complex and applied settings.

Student: Bas van Buitenen

Location: Wageningen University

Type of project: thesis

Treating malnutrition in dementia patients. The interaction between food odour exposure and cognitive ability

Dementia is the second leading cause of death in the Netherlands. The condition accounted for 12% of all elderly deaths in 2017. Dementia causes protein-energy undernutrition. Dementia patients struggle with ingesting enough nutrients to suffice their bodily needs. Previous research indicated food odour might be an effective solution to this problem. This study further evaluates the effect of food odour on appetite and nutritional intake in dementia patients. Specifically, this study aims to relate cognitive ability to the effect of food odour.

Student: Anton Wrisberg

Location: Cape Town, South Africa

Type of project: thesis

Testing the effect of behavioural science guided communications on recipients' engagement with early-stage mental health services.

Mental health related costs are the main expense of medical schemes in South Africa – an expense exaggerated by beneficiaries only seeking professional mental help at late stages. Through a literature review, the main behavioural barriers appear to be lacking knowledge of mental disorders, their symptoms, prevalence, and treatments. In other words, existing information campaigns fail. In a collaboration between MedScheme and the Behaviour Change Agency, I test whether a behavioural science guided communication strategy can improve early help-seeking amongst people (potentially) suffering from mental health disorders. This study compares MedScheme's traditional communications with behavioural science guided communications in a randomised control trial of n=2000, measuring open rates, click-through rates, and reimbursement rates.

Student: Anna-Sophie Laan

Location: Marin (maritime research institute)

Type of project: thesis

As cognitive processes are involved in effective task performance, human operators often engage in cognitive compensatory strategies to sustain outward performance. However, these cognitive processes can show evidence of strain even if outward performance is being sustained successfully and at some point, the cognitive strain will finally be accompanied by degraded performance as he or she is no longer able to cope. Therefore, measuring the operators' cognitive state in real-time can allow us to determine if the operator is and will be capable of meeting the relevant task demands and sustain performance. Assuming that we want to avoid breakdown in behavior performance, there are 'vulnerable' cognitive states that precede this breakdown - if the operator is in a vulnerable state, this suggests a period of operational risk that follows.

We want to find out what these (risk) cognitive state (or states) looks like, when it occurs and what this means for subsequent performance, and one way to do this is through physiological monitoring. We want to identify in some way the functionality of operator state rather than merely assessing different states.

This is done through EEG measurements. First, we are going to measure alpha and theta oscillations for it shows that these oscillations are sensitive to workload. Furthermore we are going to use an auditory oddball task to measure the spare capacity of the operator. Once we have mapped the workload and the influence this has on the performance of the operator, we want to find out if we can predict an overload of the operator. We will probably measure this with the alpha oscillations.

Student: Iga Jarosz

Location: ProRail

Type of project: thesis

Exploring the relationship between individual performance and non-technical skills in the system output generated in simulated railway traffic management scenarios.

Non-technical skills (NTS) are recognized as an important aspect of train traffic control (TTC), NTS complement and support the technical skills of the human operators in-the-loop. The current research focuses on exploring the possibility of assessing these skills using objective system variables generated during traffic management simulation sessions. The study has been divided into two phases i) expert knowledge elicitation through semi-structured interviews and an online survey, ii) system output analysis. Emphasis has been put on how NTS relate to the individual TTC performance variables such as train delay, unexpected train stops, or process plan changes. Presently, the first phase is underway contributing to the generation of insights that will be applied to the second phase of the research and therefore, guide the process of constructing objective NTS metrics. Furthermore, looking into patterns of activity supported by hierarchical task analysis within timeframes of subsequent events included in the scenarios will be of interest.

Student: Caterina Maidhof

Location: Utrecht University (Overvliet)

Type of project: internship

Literature Review on aspects of communication support for people relying on Augmentative and Alternative Communication (AAC)

Augmentative and alternative communication (AAC), such as signs, communication boards, speech-generating devices or mobile technology with AAC apps provide promising support for people with complex communication needs. The latter can be anyone who has not a normally functional speech and/or writing ability due to developmental or acquired disabilities. By using a literature review study, I provide a scientific overview of the aspects to consider when supporting a person relying on AAC. Support can never happen throughout a generalized approach because of people's great variability of disabilities, age ranges, environmental and social circumstances. However, aspects such as competences, communication purposes, involved stakeholders and ideal AAC assessment procedures can be generally defined. To illustrate the above, I use the example of communication support for children and young individuals, which emphasizes to value each aspect of communication support. I conclude that overall communication support must strive for the common goal of moving the impacted individual towards enhanced participation in all activities of their life.

Student: Lisa Verheul

Location: Lectoraat Co-Design, Hogeschool Utrecht.

Type of project: thesis

Parental involvement is essential for the effectiveness of the child's treatment for Developmental Language Disorder (DLD). Unfortunately, no studies have investigated parents' wishes and experiences regarding their involvement. Parents also differ in their ability and willingness to be involved. This study assessed differences in parents' experiences and wishes regarding their involvement in their child's treatment for DLD, by

using a survey and semi-structured interviews. A Thematic Network Analysis was used to give meaning to and identify parental factors. Parents were currently involved in observation of therapy sessions, completing home practice; they wished for more involvement in goal selection. Most parents learned to stimulate the child's language and received instructions to complete home practice; they wished to learn more about how to change their own language. This study gives interpretation to parents' role in the child's treatment and provides Speech and Language Therapists (SLT) with tools to negotiate roles with parents.

Student: Ziyang Liu
Location: Utrecht University
Type of project: internship

The deliverable of this internship project will be a literature review study. This study aims to investigate how oral contraceptives, via actions of their main ingredients estradiol(E) and progesterone(P), affect the development of anxiety among women. Since there are more than 100 million women use oral contraceptive pills worldwide, and some of them might encounter the side effect of anxiety. Therefore, it is essential to understand the effect of OCs on anxiety and its mechanisms underlying this process. This paper will(might) focus on 1) the fluctuation of E and P in menstrual cycle, and the effects of OCs on E and P; 2) how sex hormones influence fear conditioning and fear extinction, resulting in different outcomes of anxiety; 3) how cortisol connect to anxiety and sex hormones.

Student: Nina Geldof
Location: Utrecht University/Unilever
Type of project: thesis

At Unilever there is interest in the topic of personal care and hygiene. It is useful to know in which way odors can influence the perception of hygiene. The effect of odor on visual perception has however received little attention in research literature so far. The current study will investigate whether odor related to personal hygiene (i.e., deodorant) can enhance visual attention towards congruent visual stimuli. Participants will be asked to rate cleanliness vs. dirtiness of presented visual objects in a visual perception task. Half of them will be exposed to deodorant that is applied before the experiment. The current corona pandemic provides another angle to this study. The second aim of the study is therefore to collect information from participants on their personal hygiene rituals and product use before and during the pandemic, to determine how adhering to or dropping these rituals affects visual perception of dirt or cleanliness.

Student: Shermaily Riley
Location: TNO
Type of project: thesis

Perception of the location of the horizon in foggy circumstances for rotary aircraft pilots

Spatial disorientation training is essential for pilots. TNO is currently developing scenarios to illustrate the effects of visual illusions caused by fog, amongst others. An automated imaging processing algorithm is used to analyze the visual input for pilots. The current study explored the effect of fog on the localization of the horizon. The hypothesis was that the horizon would be perceived lower than the true horizon in foggy circumstances. The horizon informs pilots about the pitch angle of their aircraft. Participants performed a task in which they had to decide which of the two images presented, contained the lower horizon. No fog, medium fog and dense fog conditions were tested. A psychometric function was fit through the data, to investigate the bias and the discrimination threshold for the different conditions. The results indicated that fog caused a negative bias as well as a greater discrimination threshold. Furthermore, the bias of the automated imaging processing algorithm was validated.

Student: Josefine Funnekotter
Location of project: Utrecht University (Junge)
Type of project: Thesis

Due to personal circumstances and the corona crisis, my thesis has been delayed. Considering the circumstances of the corona crisis, I wanted to set up my own thesis project at the university itself. The general topic of my thesis is: the influence of language in the media. I'm currently collecting material and discussing it with my supervisor, in order to set up a research question and plan. Because the thesis project is not linked to research of a company/institute or a professor at university, it takes a bit more time to define the direction. Nevertheless, the process of setting up a research project yourself is definitely interesting and fun.

For now, the focus has been on linking the Framing Effect (Tversky & Kahneman, 1981) to language use in the media. Many parameters and a clear research question are still to be defined.

Student: Eline Nispeling
Location: Rijkswaterstaat
Type of project: thesis

Comprehensibility of the Red Cross in Different Traffic Scenarios: Results from online focus group discussions

It is commonly seen that some road users in The Netherlands chose to drive on a lane despite it being closed, as signified by a red cross. Until now, red cross negation (RCN) has often been assigned to devaluation, which refers to the tendency of road users to dissociate the red cross from danger. Yet, it cannot be ruled out that in some cases, RCN occurs unintentionally. The results of the first study suggested that a substantial amount of road users drive on the closed lane because they misinterpret the red cross, given the presence of a closed exit lane or vehicles ahead driving on the closed lane. However, it was not shown why these traffic scenarios supposedly provoke confusion regarding the red cross and its validity. In this follow-up study, a qualitative research design was chosen using online focus group discussions to gain a deeper understanding of comprehensibility of the red cross and

why it is affected in different traffic scenarios. A total of six focus group discussions were conducted.

Student: Rania Kaparelioti

Location: Signify (Philipps Lighting), Eindhoven

Type of project: internship

Smart home - Hue Lights Automation Process

Abstract: How people perceive and adopt smart home technology is a great challenge for technology companies, as people can perceive smart home technology in a wrong way and to be quite slow adopters of smart home technology. The majority of the literature regarding smart home has been mainly focused on perception of control among users and machines. We believe that one major obstacle for adopting smart home technology is the installation process of the automation software and more specifically difficulties during the setting up of the smart home technology/ hue lights. Furthermore, we will examine whether users are also aware of the capabilities of the smart light regarding automation. In the first phase, literature review regarding automation and smart home was carried out. In the second phase, we are planning to have some interactions with a power users' community of the hue lights (Reddit and Facebook hue group). We will carry out ux interviews and an online survey with the users in order to answer our study's questions. In the end, after the conclusion of the interviews and the online survey we will proceed in analysis of the data and providing the final report.

Student: Shea Gilmore

Location: ProRail

Type of project: thesis

The Future Workplace of Train Traffic Controller Abstract

In the Netherlands the railway sector is expected to experience a sharp incline in train traffic in the near future, meaning that Dutch railway traffic control rooms will need to be adapted and updated in order to cope with this change, with future traffic control room design needing both coherency and precision. Train Traffic Controllers (hereafter TTC) are the professionals that manage the control rooms, and work with complex systems in a fluctuating environment, making their profession a highly skilled and specialised one. Increasing digitisation of safety operator environments has been thought to result in lowered operator situational awareness, and uneven cognitive workload distribution because environments fluctuate between repetitive, mostly automated tasks, and high pressure situations. It has thus been proposed that TTC interface design and usability research must explore some degree of human factors, particularly situation awareness and mental workload in order to support automatic cognition and decision making in dynamic environments. Usability is a concrete approach for optimal design strategies, and is determined by the synergy between the user and tools in the context of the task and environment. The overall intention of the current thesis is to contribute to future control room design by exploring usability approaches through the construction of a conceptual

framework. Likewise the current thesis will address the future changes to Dutch railway operations by exploring usability tactics in a case study context.

The current thesis will strive to investigate the feasibility of utilising human factors measures to assess control room usability in a quantitative manner. It is hypothesised that quantitative usability can be used to more effectively assess usability of control room design. The results should be useful for human factors psychologists working in the ergonomics field. Given that ProRail has previously tested a series of prototypes, the current research will build on this research by assessing usability methods and tooling, in particular quantitative tooling feasibility, as well as adding more dimensionality to quantitative usability approaches, by asking 'what are the current existing methods for analysing usability in a control room?'. This appraisal of varying quantitative usability approaches will be integrated into a conceptual framework.

Student: Yongrui Zhao

Location: Neurolytics

Type of project: thesis

People are one of the most valuable assets of companies, but hiring employees is one of the most significant problems and most costly procedures in organizations, especially when employees turn out not to fit the job or organization very well. By using facial biomarkers data and self-report evaluations of individual working motivation and confidence, we can explore how well physiological signals can predict interview-related belief and which features matter the most, thus help the company making a better decision on recruitment. A burgeoning methodology, commonly referred as Machine Learning, has been proved to be powerful in improving performance on prediction tasks with larger feature size. The goal of this article is to expand the potential of machine learning approaches to interview-related belief prediction. We first introduced the concept of machine learning. Then we implemented two basic types of machine learning algorithms ---- Random Forest (RF) and Support Vector Machine (SVR), and their several variants to perform the prediction tasks and compared their performance with a relatively simple linear ordinary least squares model and ridge regression model. At last, we evaluated the potential of other machine learning algorithms with respect to interview-related belief prediction tasks.

Student: Jonas Kamps

Location: Utrecht University (Keizer)

Type of project: thesis

Under Lockdown: Social Distancing, Longing for Touch and Quality of Life

All of our living situations have changed in different ways after the recent outbreak of Covid-19. I joined an Utrecht University project on psychological consequences of social distancing that started at the beginning of the current containment measures. The online questionnaire we set up measures longing for touch, perception of social touch, anxiety, tolerance of uncertainty, quality of life, and demographic factors including details on people's social, living and work situations. We set the project up with several students and university staff; each of us will be looking at different aspects of the data. The focus for my

thesis is the quality of life questionnaire we used, the WHOQOL-BREF. I am looking into how respondent's quality of life is associated with their living and work situations, and their longing for touch. We want to find out about the ways in which social isolation and lack of social touch influence quality of life. I helped with the initial online study setup in March. Since then I have spent time reading social touch literature, worked on recoding and cleaning up data, and started looking for correlations between our variables. A major part of my work is cleaning up the responses we have – with over 4000 respondents so far, there is a lot of incomplete or unusable data that needs recoding. I am learning about setting up online studies, working with large and partly messy datasets, social touch research, and some of the psychological outcomes of our current, unusual situation.